

SQL Interview Questions

Q1- What is SQL

Ans- Structured Query Language is a programming language used to manage **relational database**, it allows user to store, manipulate & retrieve the data from database.

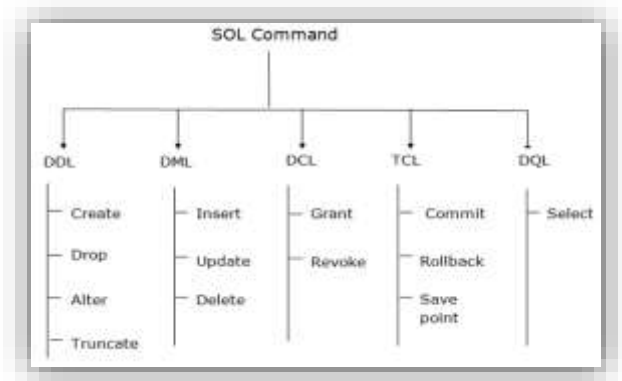
Q2- What does mean of DBMS & RDBMS, deference in between?

DBMS- DBMS is a software, used to define, create & maintain the database. In DBMS, the data is stored as a file format.

RDBMS- RDBMS is the acronym for the Relational Database Management system, whereas data is stored in the form of tables.

<u>RDBMS</u>	<u>DBMS</u>
Data stored is in table format	Data stored is in the file format
Multiple data accessible together	Individual access
Data in the form of a table are linked together	No connection between data
Data is stored in a large amount	Data stored is a small quantity
RDBMS supports multiple users	DBMS supports a single user
Oracle, SQL Server.	XML, Microsoft Access.

Q3 Types of Commands in SQL Server.



1. **DDL** – Data Definition Language
2. **DML** – Data Manipulation Language
3. **DCL** – Data Control Language
4. **TCL** – Transaction Control Language
5. **DQL** – Data Query Language

Note: - All commands of DDL are auto committed, means all changes permanently saved in the database.

Note: - All commands of DML are **not** auto committed.

Note: - TCL commands can only be used with DML commands like INSERT, DELETE & UPDATE

Q5-What is the difference between DELETE and TRUNCATE statements?

<u>DELETE</u>	<u>TRUNCATE</u>
Delete command is used to delete a row in a table.	Truncate is used to delete all the rows from a table.
You can rollback data after using delete statement.	You cannot rollback data.
It is a DML command.	It is a DDL command.
It is slower than truncate statement.	It is faster.

Q6- What is the difference between DROP and TRUNCATE commands?

DROP command removes a table and it cannot be rolled back from the database.

TRUNCATE command removes all the rows from the table, and it also cannot be rolled back from the database.

Q29- Are NULL values same as zero or a blank space?

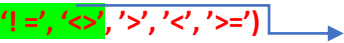
A **NULL** value is not same as **zero** or a **blank space**. NULL value represents a value which is unavailable, unknown or not applicable whereas a zero is a number and **blank space is a character**.

Q10- What do you mean by ALIAS?

Ans- This is used to temp rename a table / column name.

Q7- What does mean of Operators?

ANS- A SQL operator is a special word or character used to perform tasks.

1. Arithmetic operators, ('+', '-', '*', '/', '%')
2. Comparison operators ('=', '<=', '<', '>', '>=', '<=>')  both are same
3. Logical operators (AND, OR, NOT, All, ANY, IN, Between, EXISTS, LIKE, IS NULL)

A-Arithmetic Operators	- the arithmetic operators are used to perform mathematical operations on the numerical values stored in the database tables.
B- Comparison operators	- The Operators which compare the values of two columns in the database tables are called as comparison operators.
C- Logical operators	- The Logical operators are those that are true or false. They return a true or false values

Q8- What is the difference in both 'BETWEEN' and 'IN' condition operators?

BETWEEN operator is used to display rows based on a range of values in a row.

IN operator is used to check for values contained in a specific set of values.

Example of BETWEEN: -SELECT * FROM Students where ROLL_NO BETWEEN 10 AND 50.

Example of IN: -SELECT * FROM students where ROLL_NO NOT IN (8,15,25).

Q9- What do you mean by Like Operators?

Ans- This is mostly used for string value, where filter value is not fixed.

Exam- WHERE SALARY LIKE '_00%'

- Finds any values that have 00 in the second and third positions.

WHERE SALARY LIKE '2_%_ %'

- Finds any values that start with 2 and are at least 3 characters in length.

WHERE SALARY LIKE '%2'

- Finds any values that end with 2.

Q11- What do you mean by Union operator?

Ans- Union clause/operator used to combine the result of two or more select statements, without returning any duplicate rows. It will combine the result row wise.

A-Union- under this unique data will reflect.

B-Union All- Under this all information of both tables will reflect.

Q12- What is difference between UNION & JOIN?

JOIN	UNION
JOIN combines data from many tables based on a matched condition between them	Combines the result set of two or more SELECT statements.
It combines data into new columns.	It combines data into new rows
The number of columns selected from each table may not be the same.	The number of columns selected from each table should be the same.
Datatypes of corresponding columns selected from each table can be different.	The data types of corresponding columns selected from each table should be the same.
It may not return distinct columns.	It returns distinct rows.

Q32- What is difference between CHAR & VARCHAR/N VARCHAR?

CHAR- Is a fixed-length string data type.

VARCHAR- Is a variable-length string data type.

N-VARCHAR - It will accept alphabetic/numeric

Q13- What do you mean by Case Statement

Ans- CASE statements allow you to evaluate conditions and return a value once a condition is true.

Q14- What do you mean by Joins in SQL?

Ans - Joins are used to combine data from **two or more tables**, based on a **related column** between them.

1. **INNER Join**- Will provide common record of both tables, if any text twice in column, will provide twice in result.
2. **LEFT Join**- All data left table and common data from right table.
3. **RIGHT Join**- All data from Right Table & common data from left table.
4. **FUL Join**- Under this all record from both or all tables will appear.(common ID record will come in same lines)
5. **SELF-Join**- Under this Joins happen in same table to get the results.

NATURAL Join- Natural Join is a type of INNER JOIN that automatically joins two tables based on columns with the same name and data type. It returns only the rows where the values exist in the common columns match.

- It returns rows where the values in these common columns are the same in both tables.
- Common columns appear only once in the result, even if they exist in both tables
- Unlike a CROSS JOIN, which creates all possible combinations of rows, a Natural Join only includes rows with matching values

Note- This join is not supported by SQL Server.

Q15- What is the difference between cross join and natural join?

Cross join provides the cross product whereas Natural join will provide common record of both tables.

Q18-SQL Clause

- ❖ **GROUP BY Clause** (This comes with aggregate function)
 - If any column inserted without aggregation need to add in **group by**, line else use of aggregation
 - If multi columns added in 1st line, but any column is not aggregated, it will throw error, **use Group By there or use aggregate function with that column.**
- ❖ **ORDER BY Clause** This is used for ascending/descending, by default it is ascending.
- ❖ **HAVING Clause** **This will come with Group by (after this), without it this will not work**
 - **This works on column or aggregated column only.**
- ❖ **WHERE Clause** **This works on records** this comes before **group by**
- ❖ **Top**- This will work on **top or bottom in table record** (Top 5), 50 percent record.
 - **This will be written in select line.**

Q19- What is difference between Group by & having?

Ans- **Group by** used to group rows based on one or more columns, while **having** is used to filter grouped rows based on specific condition.

Q20- What is the difference between 'HAVING' CLAUSE and a 'WHERE' CLAUSE?

1-WHERE can be used with SELECT, Insert and Update statement. Whereas **HAVING** clause can only be used with the Select Statement

2-WHERE filters row before aggregation (GROUPING), whereas **HAVING** filters groups, after the aggregations are performed.

3-Aggregate function cannot be used in the **WHERE clause**, whereas aggregate functions can be used in **Having clause**.

Q21- What is order of execution in SQL?

Ans- Hierarchy of using/works of clause: -

- 1) **FROM**- Choose table from base data.
- 2) **WHERE**- Filter the base data.
- 3) **Group by**- Aggregate the base data.
- 4) **HAVING**- Filter the aggregated data.
- 5) **SELECT**- Return the final data.
- 6) **ORDER BY**- Sort the final data.
- 7) **Top**- Will return the nth value (Will write in select line)

Q35- Write a SQL query to display the current date?

Ans- In SQL, there is a built-in function called **select Get Date ()** which helps to return the current timestamp/date.

Q23- How can you fetch only dup record from the table

```
Select DEPTID, COUNT(DEPTID) AS CNT from EMPLOYEE  
GROUP BY DEPTID  
HAVING COUNT(DEPTID)>1
```

Q53- How to filter NULL values in the table

```
select * from batch3  
where Raj IS NULL
```

Q44- What is a "TRIGGER" in SQL?

Ans- A trigger is a special type of stored procedure that automatically runs when an event occurs in the database server.

Q30- What is a Stored Procedure and types?

Ans- A store procedure is a set of statements with an **assigned name**, which is stored in database, so it can be **re-used** and **shared** by multiple programs.

Types of Stored Procedure: -

1. **System Defined** - store procedure, exam- `sp_rename`, `sp_renamedb`, `sp_help`
2. **User Defined** - store procedure, exam- created by user

Q31- Advantages and disadvantages of Stored Procedure?

A- Advantages-

- ❖ **Reusable**- A Stored Procedure can be creating once and call for several times whenever it is required.
- ❖ **Improved performance**
- ❖ **Easy to modify**: You can quickly change the statements in a stored procedure as and when you want.
- ❖ **Time efficiency**: Once you create no need to write same query again & again.

B-Disadvantages-

The only disadvantage of Stored Procedure is that it can be executed only in the database and utilizes more memory in the database server.

Q17. What is the need for group functions in SQL?

Ans- Group functions work on the set of rows and returns one result per group. Some of the commonly used group functions are: AVG, COUNT, MAX, MIN, SUM, VARIANCE.

Q33- What is Function

Ans- functions are pre-built actions that perform calculations, manipulate, and return results. At the most fundamental level, these functions simplify to complex queries and automate particular tasks.

Types of Function: -

- 1- System Defined Function
- 2- User Defined Function

1- System Defined function - Functions that are defined by the system are known as system functions all the **built-in** functions as below: -

- Aggregate function- `SUM`, `COUNT`, `MIN`, `MAX`
- Date/Time function- `GETDATE`, `current_timestamp`, `YEAR`, `MONTH`, `DAY`
- String Function- `CONCAT`, `REPLACE`, `left`, `Right`, `Len`
- Window function- `ROW_NUMBER()`, `DENSE_RANK()`, `RANK()`

2- User Defined Function - Functions that are created by the user, called user-defined functions (**UDF**)

- **Scalar/single valued Functions**: - where functions return a single value of a specific data type.
- **Table-Valued Functions**: - where function return a table as result.

Sn.	Function	Procedure
1.	Functions always return a value or result	The procedure does not return a value, It will return result
2.	A Function cannot return <u>multiple result</u> sets.	A procedure can return multiple result sets.
3.	The function can be called using Stored Procedure.	While procedures cannot be called from function.
4.	A function used only to read data.	A procedure can be used to read and modify data.
5.	A function can be viewed by SELECT statement .	While it can be viewed by Execute statement .

6.	Function works with <u>input parameter</u>	procedure works with both parameter input-output
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Q25- How can you insert NULL values in a column while inserting the data?

Ans- Updating by specifying NULL keyword in the VALUES clause.

Q26- How can you select unique records from a table?

You can select unique records from a table by using the DISTINCT keyword.

Select **DISTINCT** studentID from Student

Q22- How can you fetch common records from two tables?

Ans- You can fetch common records from two tables using **INTERSECT**.

Q24- Deference between INTERSECT & INNER JOIN

INNER JOIN	INTERSECT
We can get new type of row	Return only those rows having common in both
Inner join, return dup records	Only unique rows reflect
Inner join, not required same order of column	No. of column, order of column should be same

Q34- What do you mean of Constraints?

Ans- Constraints are used to apply certain conditions or restrictions on the database. This ensures the accuracy and reliability of the data.

Types of Constraints: -

1- Not Null- Ensure that A column cannot have null value.

2- Default Key- Provides a default value for a column when none is, specified

3- Unique Key- Ensure all values in a column are different No Dup in that column, can accept 1 NULL

4- Primary Key- Ensure all values in a column are different No Dup in that column, can't accept NULL

5- Foreign Key- A FOREIGN KEY is a field in one table, that refers to the PRIMARY KEY in another table

6- Check- CHECK constraint on a column it will allow only certain values for this column.

Q36- What is a UNIQUE constraint?

Ans- A UNIQUE constraint is a column that uniquely identifies each row in the table, can accept 1 NULL, multiple UNIQUE KEY can be used in one table.

Q37- What is a PRIMARY KEY?

Ans- A PRIMARY KEY is a column that uniquely identifies each row in the table, cannot accept NULL value and only one PRIMARY KEY can be used in a table.

Q38- What is a FOREIGN KEY?

Ans- A foreign key is a column that connects to the primary key in the original table.

Q41- Difference between Primary Key & Unique Key.

PRIMARY KEY	UNIQUE KEY
Cannot accept NULL values.	Can accept NULL values.
Only one primary key	More than one unique key
The primary key, by default, creates clustered index	Creates non-clustered index
A Primary key supports auto-increment value.	The unique key, by default, creates a non-clustered index.
We cannot change or delete the primary key values.	We can modify the unique key column values.

Q40- What does mean of sub query?

Ans- A subquery/inner query/nested query is a query statement that appears **inside** another query statement.

Q42- What is a co-related sub query?

Ans- A Correlated sub query is a sub query that references from the outer query. it dependents on the outer query's result.

Q43- What is difference between Co-related subquery & Nested subquery?

Ans- A co-related subquery is a subquery that references from the outer query, while a nested subquery is a subquery that is independent of the outer query.

Q46- What is CTE? Use / benefit of CTE?

Ans- A Common Table Expression (**CTE**) in SQL is a one-time result set. it exists only during the execution. It allows to work within that query, such as **using it** in SELECT, INSERT, UPDATE, DELETE.

Q- Difference between Table & view.

Table- Table is a collection of rows & columns and related data entries. Table is a primary storage for storing data and in database, in table can insert, delete or update the data.

View- View is a virtual table. The result looks like a table; however, this table is not physically stored in the database.

Due to some confidentiality we don't want to share all information to user, hence we create a view table with required information, in View we can't insert, delete or update the data.

Q47- What is ACID property in a database?

Ans- Its group of SQL statement, which is considered as single unit of task, for Example-- if any transaction happen then transaction will be or not completed by any reason, if not completed, it will be rollback.

Properties of Transaction: -

- **Atomicity:** - Ensures that all operations within the unit are completed successfully, otherwise the transaction is aborted at the time of failure, and previous operations are rolled backed. (All or nothing) means all statement will successful or all fail.
- **Consistency:** - Ensures that database properly changes state, upon a successfully committed transaction.
- **Isolation:** - Enable transactions to operate independently of and transparent to each other (in case of multi transaction on same time)
- **Durability:** - Ensures that the result or effect of a committed transaction persist in case of system failure. (Information should be forever in the database)

Q48- What Is a Composite Key in SQL?

Ans- A composite can be defined as a combination of multiple columns, and these columns are used to identify all the rows that are involved uniquely.

Q50- What is Candidate key?

Ans- A candidate key is a column or a combination of columns that uniquely identifies each row in a table.

- Candidate Key can be single or multiple in a table
- Any field or column name can be Candidate Key
- It must contain unique values

Q51- What is Index?

Ans- Indexes Are Special type of Data Structure associated with tables or views that helps to speed up the query and quick data retrieval from the database.

Q52- What are clustered and non-clustered index in SQL?

Clustered Index- A clustered index defines the physical order in which table records are stored in a database, by default clustered index is created on a primary key column.

Non-clustered Index- a non-clustered index does not determine the physical order of data and this is stored in a separate location from the data table.

CLUSTERED INDEX	NON-CLUSTERED INDEX
A clustered index is faster.	A non-clustered index is slower.
The clustered index requires less memory.	A non-Clustered index requires more memory.
A table can have only one clustered index.	A table can have multiple non-clustered indexes.
The size of The primary clustered index is large.	The size of the non-clustered index is compared relatively The composite is smaller.
By default a clustered index is created on a primary key column.	It can be used with unique constraint on the table which acts as a composite key.

Q54- Difference among Rank (), Dense Rank() & Row Number().

Rank- Rank function used to assign a rank to each row, if two or more rows have the same values, they will receive the same rank, but for next row, it will skip the rank.

Dense Rank- Dense Rank function used to assign a rank to each row, if two or more rows have the same values, they will receive the same rank, but for next row, it will not skip the rank.

Row Number- Row Number function used to assign a unique sequential number to each row, if used partition by then again show unique number for each part

Value	Dense rank	Rank	Row number
10	1	1	1
20	2	2	2
20	2	2	3
30	3	4	4

Q55- Why Group by not used with Window Function?

Ans- Group by will **not** work under Window function,

If you were to use **GROUP BY** department-id, you would only get one row per department-ID, showing the average salary, but you would not see **individual employee salaries** in the same result set.

GROUP BY is for summarizing data by collapsing rows into groups.

While window functions are for performing calculations over a **set of related rows** without **losing** the individual row details.

Q55- What does mean of SSMS?

Ans- SSMS is SQL Server Management Studio, this is only Client Tool, all data stored in database server and we use client tool to connect the database server.

Q56- What does mean of Visual Studio?

Ans- Before uploading data to database server, we required ETL activities **AND** we do this activity in Visual Studio, then after we upload data in SQL Database Server. As like Power query (ETL) for Power BI.

<https://www.youtube.com/watch?v=msCJxaA63IA>

For any business 3 steps are very important, all below steps comes under 3 products

- 1- Collect (Data)- This comes under **SSIS**- SQL Server Integration Services
- 2- Analysis (Data)- This comes under **SSAS**- SQL Server Analysis Services
1. Present/view (Data)- This comes under **SSRS**- SQL Server Reporting Services

https://www.youtube.com/watch?v=I_Ae3suaL-U

1. **SSIS**- SSIS stands for SQL Server Integration Services. It's a Microsoft platform used for data **integration** and **transformation**. SSIS enables users to extract data from various sources and transform it according to business requirements, and load it into target destinations such as databases, data warehouses.
2. **SSAS** SSAS stands for SQL Server Analysis Services. It's a Microsoft Business Intelligence tool used to create online analytical **processing** (OLAP) and data mining solutions. SSAS enables user to **design** and **deploy** multidimensional data models
3. **SSRS**- SSRS stands for SQL Server Reporting Services. It's a server-based reporting platform developed by Microsoft, that enables user to create, management, and delivery of interactive reports. SSRS is widely used for **building** and **distributing** reports in organizations.

Q- What is Query Optimizing/how to do the query optimizing?

Ans- SQL query optimization is the process of enhancing the performance of SQL queries to **ensure** they execute as efficiently as possible. The goal of query optimization is: -

1. Minimize the resources.
2. Reduce the query execution time.

Why do optimization: -

Imagine a customer searching for a product online and results take a few minutes to appear. Would the customer wait? Probably not. For this reason, optimized is required.

Ways to Optimize SQL Queries: -

1. Use of Indexing
2. Avoid SELECT queries
3. Reduce the use of wildcard characters
4. Use appropriate data types
5. Avoid redundant or unnecessary data retrieval
6. Avoid subqueries

Q- Difference between T-SQL & PL-SQL, SQL & My SQL

Q- What is Temp Table & Global table and differences?

Temporary Table

Temp table that exist temporarily on the SQL Server, temp table stored in **TempDB**, they work like regular table in this table you can perform the operations like: - insert, update & delete command, temp table have two types.

1-Local Temp Table

Local temp tables are similar to Permanent Table; it accepts the single hash value "#" as prefix at the time of creation. They are visible only to the connection that creates it and are deleted when the connection is closed.

2-Global Temp Table

Global Temporary Tables are also similar to Local Temporary Tables; it accepts the double hash value "##" as prefix at the time of creation. They are visible to all connections of SQL Server and only destroyed when the connection last referencing the table is closed (in which we have created the Global Temporary Table).

Why its required: -

A temporary SQL table, also known as a temp table, is a table that is created and used within the context of a specific session or transaction in a database management system. It is designed to store temporary data that is needed for a short duration and does not require a permanent storage solution.

How to create-

```
Select * into #LocalTemp from employees-----Local Template  
Select * into ##LocalTemp from Employees-----Global Template
```

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